REVISED READING GUIDE · 7 SEPTEMBER 2026

# Forex dependence, one formula at a time

28 methods, with readable equations, symbol definitions and short explanations.

## Start with returns, not raw price levels

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

P is the exchange rate; t is the current observation; t−1 is the preceding observation. ln is the natural logarithm. Except for cointegration and explicitly identified price-increment methods, the formulas use returns.

**Example:** EUR/USD rises from 1.1000 to 1.1110. Its log return is ln(1.01) ≈ 0.009950, or 0.9950%. The ordinary percentage return is 1.0000%.

## The notation used throughout

- **x and y:** matched return series, such as EUR/USD and GBP/USD.
- **t, n, k:** date index, number of observations, and lag/horizon.
- **x̄, ∑, √:** average of x, add the terms, and square root.
- **|x|:** magnitude; for example, |−0.01| = 0.01.
- **Corr:** apply the Pearson formula in method 01.
- **Cov / variance:** joint movement around averages / squared fluctuation size.
- **1(condition):** 1 when true, otherwise 0.
- **A hat (ˆ):** an estimate; a tilde (˜): a transformed quantity.

> **How to read the guide**
>
> Read the equation, then its symbol key, then the explanation. All examples are hypothetical. A relationship can describe risk without predicting profitable trades.
>
> Keep quote directions consistent: reversing USD/JPY to JPY/USD reverses its log-return sign. Several dollar pairs also share USD exposure.

## 01 Pearson correlation

$$\rho = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum_t (x_t - \bar{x})^2 \times \sum_t (y_t - \bar{y})^2}}$$

**Symbols.** $x_t$ and $y_t$ are the two returns on date t; $\bar{x}$ and $\bar{y}$ are their averages. $\sum$ means add over all dates; $\sqrt{}$ means square root.

**How it works.** Subtract each series' average, multiply the two deviations on each date, and add them. Divide by the size of the two series' fluctuations so the answer lies between −1 and +1.

**Example.** If above-average EUR/USD returns tend to accompany above-average GBP/USD returns, Pearson tends to be positive.

**Remember.** ρ = 0.8 does not mean an 80% chance of moving together. Zero Pearson correlation can still hide nonlinear dependence.

SciPy: Pearson correlation (formula and interpretation)

## 02 Spearman rank correlation

$$\rho_s = \mathrm{Corr}(\mathrm{rank}(x), \mathrm{rank}(y))$$

**Symbols.** rank(x) replaces returns with their positions from smallest to largest. Corr means the Pearson calculation in method 01.

**How it works.** Rank each series separately, then correlate the ranks. This asks whether relatively high observations in one series accompany relatively high observations in the other, without emphasizing the size of unusually large returns.

**Example.** For x = [−2, 0, 3] and y = [−1, 1, 9], both rank lists are [1, 2, 3], so Spearman is +1.

**Remember.** Use average ranks for ties. It captures monotonic relationships, but can miss a U-shaped relationship.

SciPy: Spearman rank-order correlation

## 03 Kendall tau-b

$$\tau_\beta = \frac{C - D}{\sqrt{(C + D + T_x)(C + D + T_y)}}$$

**Symbols.** C counts concordant pairs of dates; D counts discordant pairs. $T_x$ and $T_y$ count ties only in x or only in y; pairs tied in both are excluded.

**How it works.** Compare every pair of dates. A comparison agrees if the date with the larger x return also has the larger y return. Subtract disagreements from agreements, then correct for the number of comparisons and ties.

**Example.** Without ties, 8 agreements and 2 disagreements give τ = (8 − 2)/10 = 0.6.

**Remember.** A tau of 0.6 is not numerically equivalent to a Pearson coefficient of 0.6.

SciPy: Kendall tau-b definition

## 04 Robust correlation: clipped returns

$$\tilde{x}_t = \min(c, \max(-c, x_t))$$
$$\rho clip = \mathrm{Corr}(\tilde{x}, \tilde{y})$$

**Symbols.** c is a positive return cutoff chosen before testing; $\tilde{x}_t$ is the clipped return. Apply the same rule to y. All inputs must use the same return units.

**How it works.** Limit the influence of extreme observations, then calculate Pearson. For example, c = 0.02 caps returns at ±2% in decimal units. This simplified version shows the clipping idea; the original review additionally scales by median/MAD.

**Example.** A −7% observation becomes −2% under that illustrative cutoff, so it cannot dominate the covariance as strongly.

**Remember.** This is a sensitivity check, not a universal robust estimator. Clipping discards genuine crash information; keep the raw-data result too.

Construction used in this guide; Pearson reference: SciPy

## 05 Rolling correlation

$$\rho_t^{(W)} = \mathrm{Corr}(x_{t-W+1}, \ldots, x_t ; y_{t-W+1}, \ldots, y_t)$$

**Symbols.** W is the number of observations in the window. The semicolon separates the x and y lists supplied to Corr.

**How it works.** Calculate correlation using only the latest W observations. On the next date, remove the oldest observation and add the newest. Repeat to see how the estimated relationship changes.

**Example.** W = 60 means the most recent 60 available daily returns, not 60 calendar days.

**Remember.** Short windows are noisy. A change in the rolling estimate does not by itself prove a change in the underlying relationship.

Engle (2002): Dynamic Conditional Correlation; rolling estimator context

## 06 EWMA correlation

$$c_t = \lambda c_{t-1} + (1-\lambda) x_t y_t$$
$$v_{x,t} = \lambda v_{x,t-1} + (1-\lambda) x_t^2$$
$$\rho_t = \frac{c_t}{\sqrt{v_{x,t}\, v_{y,t}}}$$

**Symbols.** $x_t$,$y_t$ here are mean-adjusted returns; c is covariance, v is variance, and $0 < \lambda < 1$ controls memory. Calculate $v_y$ just like $v_x$.

**How it works.** Update yesterday's covariance and variances by adding the latest return products. Recent observations get more weight; older ones fade gradually. Normalize covariance to obtain correlation.

**Example.** λ = 0.94 gives today's observation 6% weight and the previous estimate 94%.

**Remember.** The estimate uses today's observed return, so it can only inform a later decision. Initialize and select λ using training data.

J.P. Morgan/Reuters (1996): RiskMetrics Technical Document

## 07 Partial / residual correlation

$$x_t = a + bz_t + e_{x,t}$$
$$y_t = c + dz_t + e_{y,t}$$
$$\rho partial = \mathrm{Corr}(e_x, e_y)$$

**Symbols.** z is a control factor; a,c are intercepts; b,d are fitted factor sensitivities. $e_x$,$e_y$ are the unexplained residual returns.

**How it works.** Regress each return on the same control factor, subtract the fitted part, then correlate what remains. Multiple factors can replace z.

**Example.** Ask whether EUR/USD and GBP/USD still move together after accounting for a specified dollar-related factor.

**Remember.** The control factor's construction matters. A basket containing the target currencies can create mechanical overlap; this does not prove causal isolation.

Basnarkov et al. (2019): Correlation Patterns in Foreign Exchange Markets

## 08 Shrinkage correlation

$$\hat{\Sigma} = (1 - \delta)S + \delta\mu I$$
$$\rho_{ij} = \frac{\hat{\Sigma}_{ij}}{\sqrt{\hat{\Sigma}_{ii}\,\hat{\Sigma}_{jj}}}$$

**Symbols.** S is the sample covariance matrix; I is the identity matrix; μ is average sample variance; δ between 0 and 1 is the shrinkage weight.

**How it works.** Blend a noisy estimated covariance matrix with a simpler diagonal target. Then divide each covariance by the corresponding standard deviations to obtain correlations.

**Example.** With δ = 0.2, the covariance estimate uses 80% sample information and 20% of the simple target.

**Remember.** Shrinkage improves stability, not necessarily predictive direction. Ledoit–Wolf estimates the weight rather than assuming a universal value.

Ledoit and Wolf (2004): A well-conditioned estimator for large-dimensional covariance matrices

## 09 DCC-GARCH

$$z_{i,t} = \frac{\varepsilon_{i,t}}{\sigma_{i,t}}$$
$$q_{ij,t} = (1 - a - b)\bar{q}_{ij} + a\, z_{i,t-1} z_{j,t-1} + b\, q_{ij,t-1}$$
$$\rho_{ij,t} = \frac{q_{ij,t}}{\sqrt{q_{ii,t}\, q_{jj,t}}}$$

**Symbols.** ε is the return surprise after a mean model; σ is its GARCH volatility. z is a standardized shock; q is an intermediate covariance-like quantity; q̄ is its long-run target.

**How it works.** First model each pair's changing volatility. Then update dependence using standardized shocks and the previous dependence estimate. Finally normalize q into a correlation.

**Example.** A 1% return is a larger standardized shock in a quiet market than in a volatile market.

**Remember.** a controls response to new shocks; b controls persistence. Fit with a,b ≥ 0 and a+b < 1, and check the marginal models.

Engle (2002): Dynamic Conditional Correlation

## 10 Asymmetric DCC

$$n_{i,t} = \min(z_{i,t}, 0)$$
$$q_{ij,t} = A_{ij} + a\, z_{i,t-1} z_{j,t-1} + b\, q_{ij,t-1} + g\, n_{i,t-1} n_{j,t-1}$$

**Symbols.** z and q come from DCC; n keeps negative standardized shocks and sets positive shocks to zero. g controls the extra negative-shock effect.

**How it works.** Add a separate term for simultaneous negative shocks. The model can therefore respond differently to joint negative and joint positive moves.

**Example.** Test whether two consistently quoted currency returns become more closely linked following joint negative shocks.

**Remember.** A = (1−a−b)Q̄ − gN̄, where N̄ is the target for nn′. Enforce positive definiteness; quote inversion changes what a negative shock means.

Cappiello, Engle and Sheppard (2006): Asymmetric Dynamics in the Correlations of Global Equity and Bond Returns

## 11 Regime-conditioned correlation

$$\rho quiet = \mathrm{Corr}(x, y \mid state = quiet)$$
$$\rho stress = \mathrm{Corr}(x, y \mid state = stress)$$

**Symbols.** The vertical bar means ‘using observations in this condition.’ State labels may be based on a prespecified trailing-volatility rule.

**How it works.** Separate observations into meaningful states, then calculate a correlation within each state. This is the easiest version to understand; a hidden-state model estimates state probabilities instead.

**Example.** Compare the relationship during low-volatility periods with high-volatility periods.

**Remember.** For trading, define states using information already available. Hidden-state mixture covariance needs between-state mean effects; do not simply average correlations.

Benediktsdóttir and Scotti (2009): Exchange Rates Dependence: What Drives It?; state motivation

## 12 Lagged cross-correlation

$$\rho X \rightarrow Y(k) = \mathrm{Corr}(x_t, y_{t+k})$$

**Symbols.** k is a positive number of periods. $x_t$ is today’s return and $y_{t+k}$ is the other return k periods later.

**How it works.** Shift the second return series forward, align the dates, then correlate. This asks whether movements in X tend to be followed by movements in Y.

**Example.** k = 1 compares today’s EUR/USD return with the next period’s GBP/USD return.

**Remember.** A lagged relationship can come from common news or mismatched timestamps. This is a screening measure, not proof of a usable forecast.

statsmodels: Cross-correlation function; this guide explicitly defines its lag direction

## 13 Granger predictability

$$y_t = c + a\,y_{t-1} + b\,x_{t-1} + e_t$$
$$\text{Null hypothesis: } b = 0$$

**Symbols.** c is an intercept; a,b are fitted coefficients; e is the unexplained error. This displayed example uses one lag.

**How it works.** Compare a model using Y's own past with one also using X's past. Test whether the added coefficient helps, then check whether the gain survives on later data.

**Example.** Does yesterday's EUR/USD add information beyond yesterday's GBP/USD when predicting today's GBP/USD?

**Remember.** With several lags, test all X-lag coefficients jointly. Granger predictability does not establish economic causation.

Granger (1969); statsmodels Granger test documentation

## 14 Distance correlation

$$a_{ij} = |x_i - x_j|$$
$$A_{ij} = a_{ij} - \text{row mean} - \text{column mean} + \text{overall mean}$$
$$\text{dCor} = \sqrt{\frac{\text{mean}(A_{ij}\,B_{ij})}{\sqrt{\text{mean}(A_{ij}^{\,2})\,\text{mean}(B_{ij}^{\,2})}}}$$

**Symbols.** A is the double-centered distance matrix for x; B is built identically for y. mean averages across all i,j combinations.

**How it works.** Measure how far apart every pair of observations is in each series. Remove average-distance effects, then check whether large distances in one series accompany large distances in the other.

**Example.** It can detect a relationship like y = x² that ordinary Pearson correlation may miss.

**Remember.** This is the usual biased sample version. It is unsigned, between 0 and 1; small positive sample values do not automatically establish dependence.

Székely, Rizzo and Bakirov (2007): Measuring and testing dependence by correlation of distances

## 15 Mutual information

$$I(X;Y) = \sum_a \sum_\beta p_{a\beta} \ln \frac{p_{a\beta}}{p_a p_\beta}$$

**Symbols.** a,b label bins; $p_{a\beta}$ is the joint-bin probability; $p_a$,$p_\beta$ are the separate probabilities. ln is the natural logarithm.

**How it works.** Compare how often two return categories occur together with how often they would occur under independence. Add the probability-weighted log ratios.

**Example.** If large moves in EUR/USD make certain GBP/USD outcomes less surprising, the series share information.

**Remember.** This is a discrete/binned formula; continuous returns can use KSG estimation. Binning affects results. MI is unsigned, not restricted to 1, and natural logs give nats.

Kraskov, Stögbauer and Grassberger (2004): Estimating mutual information

## 16 HSIC: kernel dependence

$$K_{ij} = \exp\left(\frac{-(x_i - x_j)^2}{2\ell_x^{\,2}}\right)$$

$$\mathrm{HSIC} = \frac{\mathrm{tr}(KHLH)}{n^2}$$

**Symbols.** K measures similarity among x observations; L does the same for y. ℓ is a bandwidth; H centers the matrices; tr adds diagonal entries; n is sample size.

**How it works.** Turn each series into a map of which observations resemble each other. Center the maps and measure how strongly they agree, including nonlinear patterns.

**Example.** It can flag dependence even when there is no straight-line or rank-monotonic relationship.

**Remember.** H = I − 11′/n. This is the biased estimator; it is unsigned and bandwidth-dependent. Use time-series-valid significance tests.

Gretton et al. (2007): A Kernel Statistical Test of Independence

## 17 Transfer entropy

$$TEX{\rightarrow}Y = I(Y_{t+1}\ ;\ Xpast \mid Ypast)$$

**Symbols.** I is conditional mutual information. Xpast and Ypast are chosen lag histories; the bar means ‘after accounting for.’

**How it works.** Ask how much X’s past reduces uncertainty about Y’s next value after Y’s own past is already known. It is an information-based, potentially nonlinear predictive measure.

**Example.** Does past EUR/USD add information about future GBP/USD beyond GBP/USD’s own history?

**Remember.** History length and estimation bias matter. Shared omitted drivers can create apparent information flow; it is not structural causation.

Schreiber (2000): Measuring Information Transfer

## 18 Copulas

$$u_t = F_x(x_t),\ v_t = F_\gamma(y_t)$$
$$F_{x\gamma}(x,y) = C(F_x(x),\ F_\gamma(y))$$

**Symbols.** $F_x$ and $F_\gamma$ turn each return into its percentile; $F_{x\gamma}$ is the joint cumulative distribution; C is the copula joining them.

**How it works.** Model each currency’s return distribution separately, convert observations to percentiles, then model how the percentiles occur together. A dynamic copula lets its parameters change over time.

**Example.** Ask whether the worst 5% of EUR/USD outcomes tend to coincide with unusually bad GBP/USD outcomes.

**Remember.** A copula is a distribution model, not one universal correlation formula. Different copulas imply different tail behavior.

Patton (2006): Modelling Asymmetric Exchange Rate Dependence

## 19 Tail dependence

$$\hat{L}(\alpha) = \frac{\text{number of dates with } u \le \alpha \text{ AND } v \le \alpha}{\text{number of dates with } v \le \alpha}$$
$$\lambda L = \lim_{\alpha \to 0} P(u \le \alpha \mid v \le \alpha)$$

**Symbols.** u,v are return percentiles; α is a small cutoff such as 0.05. $\hat{L}$ is a finite-threshold estimate; λL is the theoretical extreme-tail limit.

**How it works.** Count how often both series are in their lower tails, then divide by the number of lower-tail events in the conditioning series. Repeat for upper or opposite tails when relevant.

**Example.** If 12 of 20 GBP/USD lower-tail dates also have EUR/USD in its lower tail, the illustrative conditional frequency is 12/20 = 0.60.

**Remember.** 0.60 at a 5% cutoff is not an estimate with unlimited precision and is not identical to the asymptotic limit. Report event counts.

Patton (2006): Modelling Asymmetric Exchange Rate Dependence; copula tail framework

## 20 Cross-quantilogram

$$h_{x,t} = 1(x_t < q_x) - \tau_x$$
$$\rho\tau(k) = \frac{\sum_t h_{x,t}\, h_{\gamma,t+k}}{\sqrt{\sum_t h_{x,t}^{\,2} \times \sum_t h_{\gamma,t+k}^{\,2}}}$$

**Symbols.** $q_x$ is the $\tau_x$ quantile; 1(condition) equals 1 if true and 0 otherwise. Build $h_\gamma$ in the same way. k is the lead horizon.

**How it works.** Convert returns into centered quantile-hit indicators. Compare today's event in X with a later event in Y using a normalized cross-product.

**Example.** Does a bottom-5% EUR/USD return tend to precede a bottom-5% GBP/USD return one day later?

**Remember.** Unlike an all-return correlation, this targets a chosen part of the distribution. Estimate quantiles without future data and allow for serial dependence.

Han, Linton, Oka and Whang (2016): The cross-quantilogram

## 21 Spectral coherence

$$\gamma^2(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)\, S_{yy}(f)}$$

**Symbols.** f is frequency; $S_{xx}$,$S_{yy}$ measure power in each series at that frequency; $S_{xy}$ is the cross-spectrum linking them.

**How it works.** Break the series into frequencies and compare their shared behavior at each frequency. Normalize shared power by each series' own power.

**Example.** Daily data can show a relationship around a five-day period, corresponding to frequency 1/5 cycles per observation.

**Remember.** Coherence is 0 to 1 and has no positive/negative sign. Spectral estimates require smoothing; phase supplies separate timing information.

SciPy: magnitude-squared coherence using Welch's method

## 22 Wavelet correlation / coherence

$$\rho\text{wavelet}(s) = \text{Corr}(W_x(s,\cdot), W_y(s,\cdot))$$

$$R^2(s,t) = \frac{|S(s^{-1} W_x W_y^*)|^2}{S(s^{-1}|W_x|^2)\, S(s^{-1}|W_y|^2)}$$

**Symbols.** W gives wavelet coefficients; s is scale/horizon; t is time; S is smoothing; * means complex conjugate. The first formula uses real coefficients.

**How it works.** Wavelets localize fluctuations by both time and horizon. Correlation compares coefficients at a scale; coherence measures local shared strength while phase shows relative timing.

**Example.** EUR/USD and GBP/USD may be linked over multiweek horizons but weakly linked over short horizons during the same period.

**Remember.** Check edge regions and future-data leakage from two-sided filters. Coherence strength alone does not indicate direction or causation.

Grinsted, Moore and Jevrejeva (2004): Cross wavelet transform and wavelet coherence

## 23 DCCA: detrended cross-correlation

$$X_k = \sum_{t=1}^{k} (x_t - \bar{x})$$

$$\rho DCCA(s) = \frac{\text{mean}(\tilde{X}\tilde{Y})}{\sqrt{\text{mean}(\tilde{X}^2)\ \text{mean}(\tilde{Y}^2)}}$$

**Symbols.** X is the cumulative demeaned-return profile. $\tilde{X}$,$\tilde{Y}$ are profile residuals after fitting local trends in windows of size s.

**How it works.** Accumulate returns, remove local trends inside each window, then correlate the residual profiles. Repeat for different window lengths.

**Example.** Compare detrended co-movement over 20-observation versus 100-observation scales.

**Remember.** It measures a different object from raw-return Pearson. Specify trend degree, window overlap and averaging consistently.

Zebende (2011); Kwapień et al. (2015), detrended cross-correlation framework

## 24 q-dependent / multifractal DCCA

$$G_{xy}(q,s) = \text{mean}_v[\text{sign}(c_{xy,v})|c_{xy,v}|^{\frac{q}{2}}]$$

$$\rho q(s) = \frac{G_{xy}(q,s)}{\sqrt{G_{xx}(q,s)\ G_{yy}(q,s)}}$$

**Symbols.** $c_{xy,v}$ is local detrended covariance in window v; q > 0 controls amplitude emphasis. $G_{xx}$ and $G_{yy}$ use local variances with the same power.

**How it works.** Raise the magnitude of each local covariance to q/2, retain its sign, average, then normalize. Changing q changes how strongly large fluctuations affect the result.

**Example.** q = 4 emphasizes large fluctuations more than q = 2; q = 2 recovers the corresponding DCCA coefficient.

**Remember.** Do not discard covariance signs. This guide covers positive q only; zero/negative q require separate treatment.

Kwapień, Oświęcimka and Drożdż (2015): Detrended fluctuation analysis made flexible

## 25 Volatility correlation / realized correlation

$$\rho\text{vol} = \text{Corr}(|x_t|, |y_t|)$$

$$\rho\text{realized,d} = \frac{\sum_k x(d,k)\, y(d,k)}{\sqrt{\sum_k x(d,k)^2 \times \sum_k y(d,k)^2}}$$

**Symbols.** |x| is the absolute return. For realized correlation, d is a day and k indexes synchronized intraday return intervals.

**How it works.** Volatility correlation asks whether both currencies have large moves together, regardless of direction. Realized correlation instead sums signed intraday return products and normalizes them.

**Example.** EUR/USD can rise sharply while GBP/USD falls sharply: both have high volatility, even though their return directions differ.

**Remember.** These are two related but different calculations. Intraday sampling and market noise affect realized correlation.

Hayashi and Yoshida (2005): covariance estimation and synchronization context

## 26 Hayashi–Yoshida covariance

$$\hat{C}\text{HY} = \sum_i \sum_j \Delta X_i\, \Delta Y_j\, 1(I_i \text{ overlaps } J_j)$$

**Symbols.** $\Delta X_i$ and $\Delta Y_j$ are log-price changes over observation intervals $I_i, J_j$. The indicator selects intervals with positive-duration overlap.

**How it works.** Multiply price changes whose time intervals overlap, then add them. This estimates covariance without forcing asynchronous quotes onto identical timestamps.

**Example.** If EUR/USD and GBP/USD quotes update at different instants, their overlapping return intervals can still contribute.

**Remember.** This is covariance, not directly a bounded correlation. Basic HY does not remove microstructure noise; normalizing needs suitable variance estimates.

Hayashi and Yoshida (2005): On covariance estimation of non-synchronously observed diffusion processes

## 27 Cointegration / error correction

$$e_t = \ln(PY_{,t}) - \alpha - \beta \ln(PX_{,t})$$
$$\Delta \ln(PY_{,t}) = c + \kappa e_{t-1} + \text{lag terms} + u_t$$

**Symbols.** P denotes a price level; α,β define the fitted long-run relation; e is its spread; κ is an adjustment coefficient; u is a new error.

**How it works.** Fit a relationship between log prices, then test whether the residual spread is stationary. If supported, an error-correction model describes adjustment following a deviation. Johansen generalizes this to several series.

**Example.** Two pairs may have correlated daily returns but a spread that drifts indefinitely; that is not sufficient for a mean-reversion trade.

**Remember.** Assess integration orders and use cointegration-specific critical values. Stability and costs need testing; β is not automatically a lot-size ratio.

Engle and Granger (1987); statsmodels cointegration test

## 28 Networks, PCA and connectedness

$$d_{ij} = \sqrt{2(1 - \rho_{ij})}$$
$$Rv = \lambda v$$
$$TCI = \frac{100 \times \sum_{i \neq j} \tilde{\theta}_{ij}}{N}$$

**Symbols.** d is network distance. R is a correlation matrix; v,λ are a PCA eigenvector/eigenvalue. $\tilde{\theta}_{ij}$ is a row-normalized forecast-error share; N is the number of series.

**How it works.** Networks group strongly related currencies. PCA finds combinations explaining common variation. Connectedness totals the forecast-error shares attributed to other series in a fitted VAR.

**Example.** A currency cluster can reveal concentrated exposure, while a dominant component can summarize a common market factor.

**Remember.** These are three system-level tools, not one coefficient. Connectedness requires a VAR/variance-decomposition step; a network edge is not a trade signal.

Basnarkov et al. (2019); Diebold and Yilmaz (2012): directional spillovers

ONE COMPLETE CALCULATION

# Pearson, worked through

Three illustrative observations, expressed in percentage-point units. This is a teaching example, not market data.

$$x = [-1, 0, 1], y = [-2, 0, 2]$$
$$\bar{x} = 0, \bar{y} = 0$$

## 1. Add the products of deviations

The averages are zero, so the deviations equal the observations.

$$(-1)(-2) + (0)(0) + (1)(2) = 4$$

## 2. Add the squared deviations

$$\text{For x: } 1 + 0 + 1 = 2$$
$$\text{For y: } 4 + 0 + 4 = 8$$

## 3. Normalize

$$\rho = \frac{4}{\sqrt{2 \times 8}} = 1$$

Each y observation is exactly twice x, so their linear relationship is perfectly positive. Correlation ignores this difference in scale. Three observations are far too few to infer a reliable market relationship.

> **What the answer does not say**
>
> It does not mean a 100% chance of a profitable trade. It does not establish lead–lag predictability or a mean-reverting price spread.

## Scope and references

This guide simplifies the 28 entries in the existing literature review. Each method has a source link. Some advanced families are shown through representative equations, not a full estimation algorithm. The original review provides the broader bibliography and testing protocol.

Formula reference: SciPy Pearson correlation documentation